



# Evaluating Wav2Vec2-Bert for Computer-Assisted Pronunciation Training for isiZulu

Alexandra Fort<sup>1</sup>, Francis Tyers<sup>1</sup>

<sup>1</sup>Computational Linguistics, Indiana University, USA

aconeil@iu.edu, ftyers@iu.edu

## Abstract

Pronunciation training is essential for learning a second language, but tools for computer-assisted pronunciation training have only been developed for a small subset of languages. Through creating and evaluating three fine-tuned versions of Wav2Vec2-Bert, this paper investigates the performance of Wav2Vec2-Bert for detecting language learner errors in low-resource settings. The results provide insight into how the data used for fine-tuning can impact performance and a thorough analysis of erroneous predictions. The evaluation of Wav2Vec2-Bert for this task offers a case study of an under-resourced language and suggestions for how a large-language model can be used to develop pronunciation training tools in under-resourced settings.

**Index Terms:** speech recognition, computer-assisted pronunciation training, under-resourced languages

## 1. Introduction

In second language acquisition, the effectiveness of pronunciation instruction for improving learner speech is well-attested [1]. In the traditional setting, used by many without access to computer-assisted pronunciation tools (CAPT) [2], pronunciation instruction requires access to a speaker or teacher of the language, while in modern settings this may require access to a computer or smartphone. Having access to CAPT tools in the target language is especially helpful for learners who do not have access to a formal classroom or speaker [3]. CAPT also has the potential to aid those in traditional learning settings, as one-on-one pronunciation instruction is limited in language classrooms [4], and lowers the anxiety associated with pronunciation practice for students [5].

Recent research has explored how advancements in computational linguistics apply to the detection of errors in the speech of language learners in order to improve CAPT [6, 7, 8]. Research on mispronunciation detection and diagnosis (MDD) forms the basis for building CAPT tools and normally takes one of three approaches: evaluating the Goodness of Pronunciation (GOP) of the utterance, using a set of phonological rules to inform detection and diagnosis, or using an end-to-end Automatic Speech Recognition (ASR) model [9].

The method of using an end-to-end ASR model is well-suited to low-resource contexts as there are many multilingually-trained models available that can be used with little or no additional training data [10, 11, 12]. This study aims to expand on research about the effectiveness of using multilingually-trained ASR systems for CAPT in under-resourced settings by evaluating the performance of a state-of-the-art model on isiZulu. IsiZulu provides an interesting case study due to the inclusion of typologically rare phonemes, such

as the implosive, lateral fricatives and clicks, in the language's phonetic inventory. Additionally, phonemic aspiration contrasts are present, which test the model's ability to distinguish between phonemes that are in allophonic variation in some of its training languages.

The goal of this work is to evaluate one of the latest multilingual ASR models, Wav2Vec2-Bert from Meta AI, for the task of detecting phoneme errors in isiZulu language learner speech. This paper presents a thorough analysis of the errors made by the models alongside a discussion about how the data used for fine-tuning impacts these errors. The methodology used in this paper is transparent and replicable with the goal of enhancing the reproducibility of the work for other under-resourced languages. Additionally, the results aid in future research on the development of a CAPT tool for isiZulu.

## 2. Literature Review

Many works use a Wav2Vec 2.0 model for MDD on English [13, 14, 15], facilitated by the large second language (L2) speech corpora with annotation of phonetic errors, TIMIT [16] and L2-arctic [17]. [18] explore the ability of Wav2Vec 2.0 to perform MDD on these popular English L2 speech corpora and find that Wav2Vec 2.0 can successfully perform MDD in ultra-low resource settings with limited fine-tuning data. Use of Wav2Vec 2.0 for MDD has been coupled with Goodness of Pronunciation and Automatic Pronunciation Assessment, both to improve MDD through multitask learning [19] and to improve intelligibility scoring [20]. [21] use momentum pseudo-labeling to improve the performance of Wav2Vec2.0 for MDD. [22] analyze the errors produced by Wav2Vec 2.0 on the TIMIT dataset. While Wav2Vec2Phoneme can provide phonetic transcription, it is not used in this paper as the cross-lingual selection of phonemes precludes important phoneme contrasts in isiZulu, such as aspiration and nasalization.

A handful of research papers detail the development of MDD and CAPT tools for low-resource languages. [23] uses Wav2Vec 2.0 to develop a CAPT mobile application for the Finnish language and experiment with methods for reducing the spikiness of the CTC algorithm's outputs to improve the quality of feedback. [24] use XLSR-53 to create a CAPT system for the Atayal language, reporting an underperformance of the model on the typologically rare sound [ɣ]. [25] boost the performance of Wav2Vec 2.0 for Vietnamese by adding a linguistic encoder to the model. [26] uses a Māori specific RNN in the automatic assessment of pepeha speech and reports that the continuous speech used to train the model may interfere with its ability to process learner speech. This paper builds on these works by presenting the work on isiZulu as a case study with the latest Meta AI model and then adding an analysis of the cross-lingual

considerations that shape the results.

### 3. Datasets

Two open-source datasets are used for fine-tuning models: one consisting of language learner data annotated for computer-assisted pronunciation training (CAPT) in isiZulu and one providing supplementary data from native speakers in isiZulu for training and testing the models. Using both a language learner corpus and a native speaker speech corpus gives insights into how data type and quantity impact CAPT tools that use these models.

#### 3.1. IsiZulu Second Language Learner Speech Corpus

The isiZulu Second Language Learner Speech Corpus (isiZulu L2 corpus) [27] consists of recordings from 12 students of isiZulu and 3 isiZulu teachers. The students and teachers read from the same list of 831 sentences and student recordings include annotation for phoneme mispronunciations and insertions. Each phoneme in an utterance receives a score of 1 or 0, with 1 marking correct pronunciation and 0 marking incorrect. Insertions are marked using their index in the sentence. Tone errors are also annotated in this corpus, but tone falls outside of the scope of this study since it is not orthographically marked. The recordings from teachers in this corpus demonstrate the correct pronunciation of the sentences.

#### 3.2. NCHLT isiZulu Auxiliary Speech Corpus

The National Centre for Human Language Technology (NCHLT) corpus [28] is an open-source corpus that is designed to be used in the development of speech technologies for the 11 official spoken languages of South Africa, one of which is isiZulu. The section of the corpus used in this research comes from the nchlt-clean split of the dataset wherein a phone-based dynamic programming (PDP) score was used to determine the recordings with the highest reliability to the prompt.

The Zulu language split has 56 hours and 14 minutes of transcribed recordings from 210 speakers (98 Male; 112 Female). Each recording consists of a speaker reading three words. The prompts for this split were sourced from South African government websites, with the most frequent trigrams selected as prompts. The transcriptions of the recordings are in the isiZulu orthography.

## 4. Methodology

#### 4.1. Fine-tuning

Using the datasets, three separate models were fine-tuned on Wav2Vec2-BERT [11]. Wav2Vec2-Bert is the latest model from Meta AI and uses training data from 143 languages, including 67 hours of ASR data from isiZulu. Wav2Vec2-Bert is chosen due to its reported ability to perform low-resource transfer learning for any language and its inclusion of isiZulu training data. The model requires fine-tuning to be used for ASR tasks but needs a limited amount of data, which is well-suited to this paper’s setting and other low-resource settings.

The first model for this paper was fine-tuned on the 2,493 gold-standard teacher pronunciations from the isiZulu L2 Corpus, the second was fine-tuned on 41871 recordings from the training split of the NCHLT corpus, and the third was fine-tuned on both the teachers’ pronunciations from the isiZulu L2 Corpus and the training split of the NCHLT corpus. The fine-tuned

versions of these models will be referred to as L2, NCHLT, and L2NCHLT throughout the paper. The use of data from teachers in the isiZulu L2 corpus during fine-tuning is particularly interesting as it includes a relatively small set of recordings, but with a pedagogical recitation style in which the teachers are enunciating clearly in the manner they would use in the classroom.

The fine-tuning data for each model is presented in the standard isiZulu orthography to utilize the isiZulu data provided in the base model, wherein standard orthographies were used, not IPA. The same parameters were used to fine-tune each model. The description of these parameters and models are available publicly at <https://huggingface.co/aconeil>.

#### 4.2. Interpreting Model Outputs

The NCHLT test split is used to provide an evaluation metric for model comparison. However, these metrics represent the ability of the model to transcribe native-speaker speech, not second-language learner errors. To evaluate the ability of the model to transcribe second-language learner speech, transcriptions are generated for the student recordings in the isiZulu L2 Corpus. The use of characters as targets in the masked prediction task allows the model to predict primary phonemes with accompaniments, such as the voiceless aspirated bilabial stop “ph.” The fine-tuned Wav2Vec2-BERT models used in this work are not constrained to in-vocabulary words, especially with the relatively scarce training data. Once the transcriptions are obtained from each model, the target sentence and the transcription are converted into an IPA-like representation wherein each contrastive phoneme is represented by one character. The one-to-one phoneme-to-character representation facilitates alignment by minimizing instances of features of the sound (aspiration, voicing, nasalization) being interpreted as separate characters by the alignment algorithm.

The Needleman-Wunsch algorithm [29] is used to align the utterances to each other. This algorithm comes from bioinformatics and is traditionally used to align sequences of nucleotides, but can be applied to any two strings. The implementation of the algorithm used in this paper scores by using a plus one for matches and a minus one for deletions and insertions. The aligned sequences are then processed to produce a score that matches the binary format of the teacher’s feedback while extracting the indices of insertions as a list. The algorithm produces “-” for the sequences identified as insertions. Following the method used in the isiZulu L2 Corpus, sequential insertions are processed as one index, and utterance-final insertions are coded as ‘EOS.’ Table 1 shows an example of the alignment produced for the sentence “Ningawuphuzi” [niŋawup<sup>h</sup>zi] (eng:Don’t (you all) drink it).

Table 1: Alignment example with 1 phoneme error and 1 insertion error

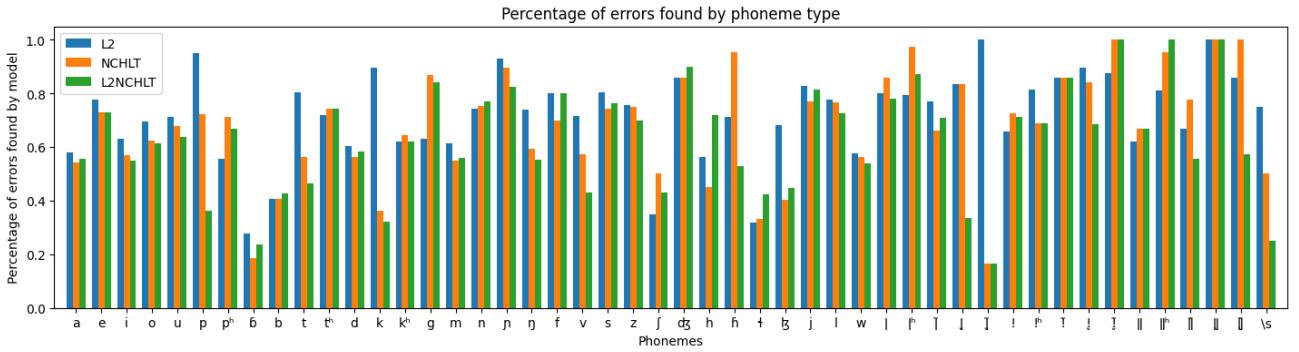
|        |     |   |   |   |   |   |   |                |   |   |   |
|--------|-----|---|---|---|---|---|---|----------------|---|---|---|
| Target | -   | n | i | ŋ | a | w | u | p <sup>h</sup> | u | z | i |
| Model  | i   | ŋ | i | ŋ | a | w | u | p <sup>h</sup> | u | z | i |
| Score  | [0] | 0 | 1 | 1 | 1 | 1 | 1 | 1              | 1 | 1 | 1 |

## 5. Results

#### 5.1. Native Speaker Speech Transcription

The results for assessment on native speaker speech below show that the model trained only on the NCHLT data performs the best. The metrics for the L2NCHLT model show that adding the teacher recordings slightly diminishes performance. The L2

Figure 1: *Percentage of errors identified for each phoneme by each model*



model does poorly when evaluated on just the NCHLT test data, likely due to the stark domain difference between the speech in the NCHLT corpus, which features naturalistic L1 speaker speech, and the recordings from the teachers, which depict careful pronunciation. All models have a much lower character error rate than word error rate, which is likely attributed to the agglutinating nature of isiZulu in which words contain more characters when compared to isolating languages.

The Wav2Vec2-Bert NCHLT and L2NCHLT fine-tuned models outperform all other models evaluated on the same test set [28, 30, 31]. As research on isiZulu ASR is limited, there are not many isiZulu ASR models to use as comparison, with even fewer having documentation. These two models perform competitively to models found on HuggingFace from asr-africa and TheirStory, with the addition of access to the code used for fine-tuning at [https://github.com/aconeil/zulu\\_mdd](https://github.com/aconeil/zulu_mdd) and access to the datasets to allow for reproducibility.

Table 2: *Model results on NCHLT test data using Word Error Rate (WER) and Character Error Rate (CER)*

| Model    | WER          | CER           |
|----------|--------------|---------------|
| L2       | 0.667        | 0.143         |
| NCHLT    | <b>0.126</b> | <b>0.0237</b> |
| L2+NCHLT | 0.141        | 0.0252        |

## 5.2. Second Language Error Transcription

This study considers phoneme deletions, mispronunciations, and insertions as labeled in the isiZulu L2 corpus, but does not consider tone, as tone is not orthographically marked in isiZulu. The binary error transcription from the teachers is used in section 5.2.1 and the insert index is used in section 5.2.2. In cases where the database has competing feedback from different teachers, the feedback that identifies more errors is chosen.

### 5.2.1. Phoneme mispronunciations and deletions

This is the first work on isiZulu pronunciation error detection, so an exploration of model performance is provided, as there is not a state-of-the-art model with which to compare. Since the gold-standard feedback marks binary errors in pronunciation without information about the phoneme produced by the learner, it is not possible to calculate the diagnostic error rate (DER) for this dataset. Instead, the true negative rate is provided. Table 3 provides the false acceptance rate (FAR), false rejection rate (FRR), and true negative rate (TNR) for each

model. FAR represents the rate at which the model failed to identify a mispronounced phoneme, while FRR represents the rate at which the model marked a phoneme as incorrect when it was pronounced correctly. TNR indicates the probability of an incorrect phoneme being identified as incorrect by the model.

Table 3: *Model results for false acceptance rate (FAR), false rejection rate (FRR), and true negative rate (TRN).*

| Model    | FAR         | FRR         | TNR          |
|----------|-------------|-------------|--------------|
| L2       | <b>2.2%</b> | 16.1%       | <b>65.1%</b> |
| NCHLT    | 3.1%        | <b>7.0%</b> | 50.8%        |
| L2+NCHLT | 3.2%        | 9.7%        | 48.6%        |

### True Negative Rate

Considering the true negative rate, the model fine-tuned on just the L2 corpus performs the best, demonstrating that it identifies the most incorrect phonemes compared to the gold standard. Figure 1 shows the percentage of mispronunciations identified by each model based on the target phoneme.

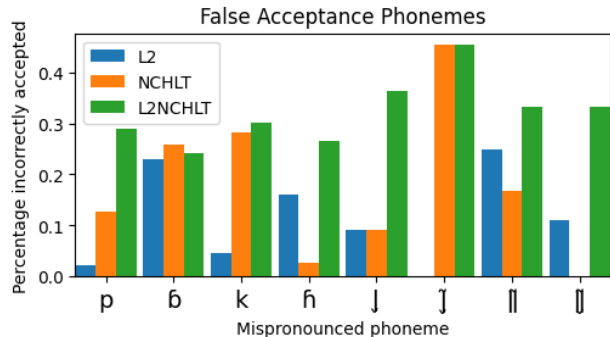
The models struggle to identify the mispronunciation of the voiced bilabial implosive [ɓ] and the lateral fricatives [ɬ] and [ɮ]. As these sounds are typologically rare, this may be due to a deficit in data for these sounds in the base model. The voiceless postalveolar fricative [ʃ] is also rarely labeled as incorrect. Additionally, we see that the careful pronunciation provided in the teachers' recordings seems to benefit the L2 model in identifying pronunciation errors for plain voiceless stops [p], [t], and [k]. For the click consonants, the models do fairly well, with some clicks showing a slight boost from the careful pronunciation in the L2 corpus ([ǀ], [ǃ], [ǁ<sup>h</sup>], and [ǂ]), while the others benefit more or equally from increased instances of pronunciation, as seen in the NCHLT corpus.

### False Acceptance Rate

Figure 2 shows the results for the 8 phonemes that were most commonly accepted by the models, despite teachers marking the phoneme as a mispronunciation. The y-axis shows the number of misidentified errors normalized by the total count of that target phoneme in the corpus. Half of these phonemes are clicks, with the NCHLT and L2NCHLT models generally missing more of these errors. Of the click consonants identified on this chart, there is representation for the co-articulation features of voicing, nasalization, and a combination of the two, suggesting detection issues may be heightened by the presence of one or more of these features. In addition to click consonants, plain stop consonants ([p] and [k]) pose a challenge to the NCHLT and L2NCHLT models, while the voiced glottal fricative [ɦ] is

frequently misidentified by the L2 and L2NCHLT model. All three models perform similarly for the bilabial voiced implosive [ɓ], which could be a result of limited instances in the base model’s training data, as [ɓ] is a typologically rare sound.

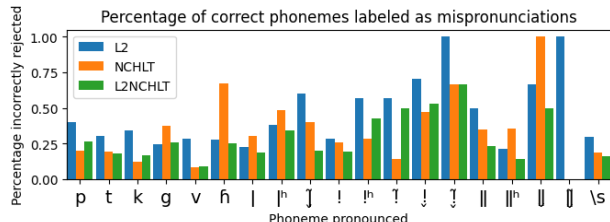
Figure 2: 8 phonemes most commonly accepted by models, despite mispronunciation.



### False Rejection Rate

Figure 3 shows phonemes that the models incorrectly labeled as errors 25% or more of the time. Despite the inclusion of careful speech in the L2 fine-tuning data, clicks are the most problematic for the L2 model. The inclusion of extra instances of clicks from the NCHLT dataset appears to minimize the number of false rejections, as the NCHLT model has fewer false rejections for 7/12 clicks. This indicates the high performance of the L2 model on the detection of click errors in 1 is at least partially attributed to the model frequently labeling these phonemes as incorrect. Further, the L2 model is more likely to falsely reject pronunciations of plain voiceless stop consonants [p], [t], and [k], which is also matched by a boost in identified true negatives in figure 1.

Figure 3: Correct phonemes rejected 25% or more of the time.



### 5.2.2. Phoneme Insertions

Of the 1,481 insertions labeled in the isiZulu L2 corpus, models were generally eager to suggest insertions, but indices often did not match those in the isiZulu L2 corpus. The NCHLT fine-tuned model identifies the most insertion errors. However, the combined L2NCHLT model has the highest precision score for the identification of insertion errors, likely benefiting from an improved tokenization ability, as demonstrated by the lower percentage of false rejections of whitespace shown in Figure 3.

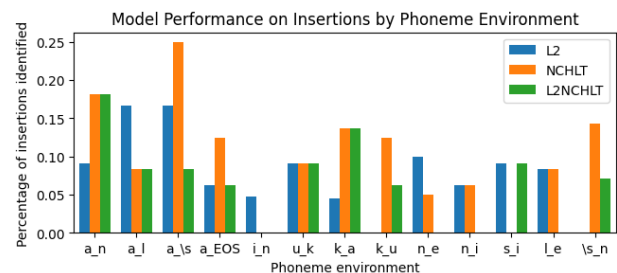
Figure 4 shows phoneme environments in which one or more of the models missed insertions 90% or more of the time. The models show varying performance for many of the environments. If the model’s bar is not showing on the graph, it indicates the model did not find any of the insertions in that environment. The NCHLT model identifies the most insertions in

Table 4: Model results for identifying sound insertions.

| Model    | Suggested | Correct    | Precision   |
|----------|-----------|------------|-------------|
| L2       | 2549      | 402        | 15.8        |
| NCHLT    | 3700      | <b>434</b> | 11.7        |
| L2+NCHLT | 2265      | 393        | <b>17.4</b> |

these challenging environments, but as shown in table 4, this comes at the expense of precision. While the analysis of the environments of missed insertions is presented to provide linguistic guidance for improving future models, further analysis of the incorrectly suggested indices is recommended to provide more context for these suggestions.

Figure 4: Insertions missed more than 90% of the time by environment. ‘\_’ represents location of insert.



The results show that the models struggle with detecting insertions between a vowel and a consonant or a consonant and a vowel, as is required by isiZulu’s strict (C)V syllable structure. These consonants and vowels do show a pattern, the vowels are [a], [i], and [u], and the consonants are either alveolar consonants ([n], [l], and [s]) or the voiceless velar stop [k]. As continuant consonants, it may be harder for the model to find the boundary between the consonant and the vowel for [s] and [l]. This may also hold true for [n] as a sonorant, but, if so, it is unclear why [m] and [ŋ] do not produce a similar challenge for the model.

## 6. Conclusion and Future Work

The results in this paper suggest that Wav2Vec2-Bert can provide a baseline for phonemic pronunciation detection in isiZulu. Evaluation on L1 speech shows that the NCHLT and L2NCHLT models exceed and outperform leading isiZulu ASR models. While more data is often considered better, fine-tuning with a smaller specialized corpus, in our case the isiZulu L2 corpus, is found to perform adequately for the task of mispronunciation detection, though with a higher tendency to overidentify errors. Suggesting that performance on standard WER and CER for native speaker speech data is not indicative of the model’s performance on mispronunciation detection for L2 speech.

Typologically rare sounds, such as [ɓ],[ɗ] and [ɰ], are challenging for all of the models, suggesting the issue is present in the base Wav2Vec2-Bert model prior to fine-tuning. However, click consonants break from this pattern and are overidentified as errors, raising the FRR for models. The models may have an easier time distinguishing clicks but fail to classify mispronunciations according to the articulatory features of the clicks. Further evaluation of the performance of Wav2Vec2-Bert for isiZulu L2 speech should include a speaker or linguist validation for the model-proposed diagnosis of mispronounced phonemes.

## 7. Acknowledgments

This research was supported in part by Lilly Endowment, Inc., through its support for the Indiana University Pervasive Technology Institute.

## 8. References

- [1] J. Lee, J. Jang, and L. Plonsky, "The effectiveness of second language pronunciation instruction: A meta-analysis," *Applied Linguistics*, vol. 36, no. 3, pp. 345–366, 2015.
- [2] P. M. Rogerson-Revell, "Computer-assisted pronunciation training (capt): Current issues and future directions," *Relc Journal*, vol. 52, no. 1, pp. 189–205, 2021.
- [3] Y.-T. Sung, K.-E. Chang, and J.-M. Yang, "How effective are mobile devices for language learning? a meta-analysis," *Educational Research Review*, vol. 16, pp. 68–84, 2015. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1747938X15000469>
- [4] J. Levis, "Computer technology in teaching and researching pronunciation," *Annual review of applied linguistics*, vol. 27, pp. 184–202, 2007.
- [5] M. Szyszka, "Pronunciation learning strategies and language anxiety," *Switzerland: Springer*, vol. 10, pp. 978–3, 2017.
- [6] K. Li, X. Qian, and H. Meng, "Mispronunciation detection and diagnosis in l2 english speech using multidistribution deep neural networks," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 25, no. 1, pp. 193–207, 2016.
- [7] M. Lounis, B. Dendani, and H. Bahi, "Mispronunciation detection and diagnosis using deep neural networks: a systematic review," *Multimedia Tools and Applications*, pp. 1–35, 2024.
- [8] B.-C. Yan and B. Chen, "End-to-end mispronunciation detection and diagnosis from raw waveforms," in *2021 29th European Signal Processing Conference (EUSIPCO)*. IEEE, 2021, pp. 61–65.
- [9] N. Alrashoudi, H. Al-Khalifa, and Y. Alotaibi, "Improving mispronunciation detection and diagnosis for non-native learners of the arabic language," *Discover Computing*, vol. 28, no. 1, p. 1, 2025.
- [10] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *International conference on machine learning*. PMLR, 2023, pp. 28 492–28 518.
- [11] L. Barrault, Y.-A. Chung, M. C. Meglioli, D. Dale, N. Dong, M. Duppenhaler, P.-A. Duquenne, B. Ellis, H. Elshahar, J. Haasheim *et al.*, "Seamless: Multilingual expressive and streaming speech translation," *arXiv preprint arXiv:2312.05187*, 2023.
- [12] A. Hannun, "Deep speech: Scaling up end-to-end speech recognition," *arXiv preprint arXiv:1412.5567*, 2014.
- [13] J. I. Kim, S. Kim, and M. Chung, "Developing a robust mispronunciation detection by data augmentation based on automatic phone annotation," in *2024 27th Conference of the Oriental COCOSDA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA)*. IEEE, 2024, pp. 1–5.
- [14] J. M. Dalhouse and K. Itou, "Cross-lingual transfer learning approach to phoneme error detection via latent phonetic representation," in *INTERSPEECH*, 2022, pp. 3133–3137.
- [15] M. A. H. Wadud, M. Alatiyyah, and M. Mridha, "Non-autoregressive end-to-end neural modeling for automatic pronunciation error detection," *Applied Sciences*, vol. 13, no. 1, p. 109, 2022.
- [16] J. S. Garofolo, L. F. Lamel, W. M. Fisher, J. G. Fiscus, and D. S. Pallett, "Darpa timit acoustic-phonetic continuous speech corpus cd-rom. nist speech disc 1-1.1," *NASA STI/Recon technical report n*, vol. 93, p. 27403, 1993.
- [17] G. Zhao, S. Sonaat, A. Silpachai, I. Lucic, E. Chukharev-Hudilainen, J. M. Levis, and R. Gutierrez-Osuna, "L2-arctic: A non-native english speech corpus," in *Interspeech*, 2018. [Online]. Available: <https://api.semanticscholar.org/CorpusID:52190513>
- [18] L. Peng, K. Fu, B. Lin, D. Ke, and J. Zhang, "A study on fine-tuning wav2vec2.0 model for the task of mispronunciation detection and diagnosis," in *Interspeech*, 2021, pp. 4448–4452.
- [19] H. Ryu, S. Kim, and M. Chung, "A joint model for pronunciation assessment and mispronunciation detection and diagnosis with multi-task learning," in *INTERSPEECH*, 2023.
- [20] R. C. Shekar, M. Yang, K. Hirschi, S. Looney, O. Kang, and J. H. Hansen, "Assessment of non-native speech intelligibility using wav2vec2-based mispronunciation detection and multi-level goodness of pronunciation transformer," *ISCA INTERSPEECH-2023*, 2023.
- [21] M. Yang, K. Hirschi, S. Looney, O. Kang, and J. Hansen, "Improving mispronunciation detection with wav2vec2-based momentum pseudo-labeling for accentedness and intelligibility assessment," in *INTERSPEECH*, 09 2022, pp. 4481–4485.
- [22] E. Loweimi, A. Carmantini, P. Bell, S. Renals, and Z. Cvetkovic, "Phonetic error analysis beyond phone error rate," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2023.
- [23] N. Phan, T. Grósz, and M. Kurimo, "Captina-a mobile app for practising finnish pronunciation," in *Nordic Conference on Computational Linguistics*. University of Tartu Library, 2023, pp. 265–270.
- [24] Y.-L. Chuang, H.-R. Hsu, Y.-W. Liu, C.-T. Hsin *et al.*, "Computer-assisted pronunciation training system for atayal, an indigenous language in taiwan," in *2024 27th Conference of the Oriental COCOSDA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA)*. IEEE, 2024, pp. 1–6.
- [25] T. Nguyen and H. D. Tran, "Lingwav2vec2: Linguistic-augmented wav2vec 2.0 for vietnamese mispronunciation detection," in *Proc. Interspeech 2024*, 2024, pp. 2355–2359.
- [26] C. I. Watson, P. Allen, P. J. Keegan, K. Mahelona, and P.-L. Jones, "Towards automatic marking of pepeha: a formulaic māori language speech," in *2nd Annual Meeting of the Special Interest Group on Under-resourced Languages 2nd Annual Meeting of the Special Interest Group on Under-resourced Languages*, 2023.
- [27] A. O'Neil, N. Hjortnaes, F. Tyers, Z. Nkosi, T. Ndlovu, Z. Mlondo, and N. P. Pewa, "Developing a benchmark for pronunciation feedback: Creation of a phonemically annotated speech corpus of isiZulu language learner speech," in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, and N. Xue, Eds. Torino, Italia: ELRA and ICCL, May 2024, pp. 4795–4801. [Online]. Available: <https://aclanthology.org/2024.lrec-main.429/>
- [28] E. Barnard, M. H. Davel, C. van Heerden, F. De Wet, and J. Badenhorst, "The nchlt speech corpus of the south african languages," in *Proc. 4th International Workshop on Spoken Language Technologies for Under-resourced Languages (SLTU)*. Workshop Spoken Language Technologies for Under-resourced Languages (SLTU), 2014.
- [29] S. B. Needleman and C. D. Wunsch, "A general method applicable to the search for similarities in the amino acid sequence of two proteins," *Journal of molecular biology*, vol. 48, no. 3, pp. 443–453, 1970.
- [30] J. Badenhorst and F. De Wet, "The usefulness of imperfect speech data for asr development in low-resource languages," *Information*, vol. 10, no. 9, p. 268, 2019.
- [31] N. Kleynhans, F. De Wet, and E. Barnard, "Unsupervised acoustic model training: comparing south african english and isizulu," in *2015 Pattern Recognition Association of South Africa and Robotics and Mechatronics International Conference (PRASA-RobMech)*. IEEE, 2015, pp. 136–141.